Saumik Dana

https://saumikdana.github.io/

Summary

US green card holder deploying end-to-end ML and LLM-based systems. PhD from UT Austin. Experience spans feature engineering, optimization, stochastic modeling, RAG, and multimodal document intelligence

Experience

Quantitative Researcher, **Zebra Capital Management LLC**, Stamford CT, **Feb 2024 - Dec 2025**

`Cloud & Infrastructure`

- Built serverless systems on AWS Lambda, CloudFormation, EventBridge, DynamoDB, S3, Modal
- Engineered GitHub CI/CD: Docker buildx+ECR registry-cache with CloudFormation deploys
- Implemented React dashboards over DynamoDB/S3 state and ECR image lifecycle cleanup

`Production ML Systems`

- Implemented anomaly detection backed volatility arbitrage intraday options trading strategies
- Deployed LoRA tuned Chronos embeddings and TabPFN backed interday options trading
- Built bi-objective optimizer backed thematic equity portfolio with regime-adaptive rebalancing

`LLM-backed Systems Experimentation`

- Evaluated Groq-backed LLM agents with RSS/XML ingestion, guardrails, and trading signal outputs
- Engineered FX and macro news driven intraday FX trading signal evaluation agents
- Compared hand-rolled and LangGraph-backed LLM architectures for FX trade position sizing
- Built ticker alert pipeline with SQLite FTS5, PostgreSQL tsvector, Whoosh, FAISS retrieval based RAG

`ML Model Development & Experimentation`

- Bi-objective optimization, A/B testing SHAP v native feature importance, Black Litterman model
- Fast jump diffusion aware Heston model parameter calibration via constrained optimization
- OOS benchmarks for Google TimesFM vs. Amazon Chronos, LoRA-adapted vs. base Chronos

`LLM-backed Research & Discovery Agents`

- Designed query-based signal mining Streamlit app over backtested options trading strategies
- Engineered multimodal embedding model and Next.js based PDF Q&A app

Computational Engineer, **VISIE Inc.**, Austin TX, **Aug 2023 - Nov 2023**

- Joined during early integration of imaging and robotic actuation in surgical navigation platform
- Implemented TCP/UDP communication protocols to control robotic arm motion
- Supported product packaging, deployment, live demonstration used in successful Series A round

Computational Lead, **Sophelio**, Austin TX, **Aug 2022 - Mar 2023**

- Adapted physics-informed modeling—originally for fusion experiment data—to financial time series
- Engineered production pipeline: sparse regression for PDE construction, signal generation
- Implemented CAGR maximizing Bayesian TPE optimizer based swing trading system

Education

Doctor of Philosophy in Engineering Mechanics, **University of Texas at Austin**, Austin TX
Master of Engineering in Mechanical Engineering, **Indian Institute of Science**, Bangalore, India
Bachelor of Engineering in Mechanical Engineering, **University of Mumbai**, Mumbai, India

Technical Skills

`Languages/UI:` Python, JavaScript, HTML, React/JSX
`Cloud:` AWS Lambda, DynamoDB, S3, ECR, EventBridge, Modal, CloudFormation, Docker, GitHub CI
`APIs/Dashboards:` Alpaca, OANDA, yfinance, Schwab, FastAPI, Mangum, Recharts, Tailwind CDN
`ML/Quant:` PyTorch, TabPFN, Chronos, PEFT/LoRA, LightGBM, HMM, pymoo, SciPy, autograd
`LLM/RAG:` Groq, LangChain, LangGraph, FAISS, Whoosh, SQLite FTS5, PostgreSQL tsvector
`Data:` pandas, NumPy, PyArrow, scikit-learn, requests, BeautifulSoup, RSS/XML, YAML